

Human-Data Interaction, Exploration & Visualization in the AI Era: Challenges & Opportunities

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Problem

AI-era data analysis now spans **large-scale, heterogeneous, multimodal, and predominantly unstructured** data — text, images, audio, video, and visually rich documents — while foundation models (LLMs, VLMs) inject **uncertainty** into every step.

This breaks the schema-driven query-then-explore workflows systems were built on: data is weakly indexed and costly to inspect.

Users must iteratively explore, reformulate, and verify results under uncertainty — often without knowing what the data contains or which questions can be meaningfully asked.

Motivation

- **Perceptual latency is a hard constraint.** Delays of even a few seconds disrupt reasoning, bias exploration, and reduce the effectiveness of human-AI collaboration.
- **Foundation models are double-edged.** LLMs and VLMs add non-determinism, error propagation, bias, and verification cost — reinforcing the need to keep humans meaningfully in the loop.

DATA SYSTEMS TODAY

VS.

HUMAN COGNITION

Tuned for **order-of-second** responses.

Needs **order-of-millisecond** feedback.

Meeting these demands requires interface-, perception-, and context-aware systems, with explicit support for oversight, provenance, and trust calibration.

Method

REFRAME

Stop optimizing data management, AI, interfaces, and visualization in isolation — **co-design them as one end-to-end, human-centered pipeline** for iterative, guidance-driven, mixed-initiative exploration.

Four coupled perspectives trace the loop: how interfaces shape systems, millisecond querying, scalable analytics, and AI-guided exploration with generative visualization.

- 1 Interface-System Co-Design
- Treat visualization as a **SQL-workload generator**: interactions become queries, so faster execution improves responsiveness.
- But P95 latency and throughput don't equal fluid UX — the interface's *own* constraints motivate new data-management problems.

System design can also **inspire** new interfaces, not merely react.

- 2 Queries at the Speed of Thought
- ML shifted from model-centric to **data-centric** — the bottleneck is now understanding data, not tweaking models.

Engineers slice and dice millions of records to grasp subtle patterns, and visualization allocates scarce attention — but only if response times match human cognition. Data systems tuned for order-of-second responses distort exploration; approximation, indexing, and prefetching buy interactivity at a cost in complexity.

- 3 Scalable Visual Analytics
- ~2% of IEEE VIS papers use a scalability keyword
- Progressive visual analytics and approximate query processing help, and stacks like Mosaic abstract low-level optimizations — but a single consensual measure of “scalable visualization” is still missing.

- 4 AI-Guided Exploration & Generative Vis
- Unstructured, metadata-poor data demands **ad-hoc, continuous query reformulation**; LLMs extract structure but add non-determinism, cost, and bias.
- A trained **GNN** optimizes arbitrary differentiable layout objectives on unseen graphs; a **GAN** optimizes non-smooth objectives from only pairwise “which is better” comparisons.

This opens a path to modeling **human aesthetic preference** directly — visualization as an active, narrative, adaptive component.

Headline Numbers

~2% of IEEE VIS papers use a scalability keyword

far below sibling data-science domains

ms latency target vs. order-of-second systems

11 authors across 10+ institutions

6 coordinated section perspectives

4 surveyed research literatures

Key Results

DIMENSION	LEGACY VIEW	AI-ERA DEMAND
Latency	order-of-second	order-of-millisecond
Data	structured tables	multimodal, unstructured
Visualization	static output	active, generative
Human role	operator	in-the-loop overseer

Only ~2% of IEEE VIS papers engage scalability — a fundamental barrier vs. sibling data-science domains.

SO WHAT → Deep learning even reshapes visualization itself: GNN/GAN models learn layout from data.

Takeaway

AI-driven ecosystems are an **inflection point**: interactive systems can no longer be built by independently optimizing data management, AI, and interaction — they must be **co-designed end-to-end**, with visualization shifting from passive display to an active, human-guided role.

Databases Foundation Models InfoVis HCI

Comp. Graphics Cognitive Science

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Paper